Introduction

Mushrooms are the fruiting bodies of fungi and constitute a significant food source. Many species of wild or field mushrooms are considered delicacies in some cultures/countries. However, there are numerous mushrooms that may prove toxic if eaten and distinguishing between "edible" and poisonous species is often difficult. In addition, some "edible" species can prove toxic to some humans in certain circumstances which may not be predictable.(White *et al.* 2019)

Mushrooms are fungi with significant nutritional value currently counting around 2000 edible species distributed around the world (Rathore et al., 2019). The most cultivated species include button mushroom (*Agaricus bisporus*), shiitake mushroom (*Lentinula edodes*), and oyster mushrooms (*Pleurotus* spp.). The current global market value of fresh mushroom reached 38 billion US dollars in 2018, and China is the largest mushroom producer within the Asian region, contributing approximately 35 % to the global mushroom market (Elaine and Tan, 2009). Asia countries contribute up to 76 % of mushroom production, followed by Europe (17.2 %) and United States (5.9 %) (Sande et al., 2019).(Mahari *et al.* 2020)

Wild mushrooms are a vital source of income and nutrition for many poor communities and of value to recreational foragers. Literature relating to the edibility of mushroom species continues to expand, driven by an increasing demand for wild mushrooms, a wider interest in foraging, and the study of traditional foods. Although numerous case reports have been published on edible mushrooms, doubt and confusion persist regarding which species are safe and suitable to consume. Case reports often differ, and the evidence supporting the stated properties of mushrooms can be incomplete or ambiguous. The need for greater clarity on edible species is further underlined by increases in mushroom-related poisonings. We propose a system for categorizing mushroom species and assigning a final edibility status. (Li *et al.* 2021)

Mushroom-forming fungi (Agaricomycetes) have the greatest morphological diversity and complexity of any group of fungi. They have radiated into most niches and fulfil diverse roles in the ecosystem, including wood decomposers, pathogens or mycorrhizal mutualists. Despite the importance of mushroom-forming fungi, large-scale patterns of their evolutionary history are poorly known, in part due to the lack of a comprehensive and dated molecular phylogeny. Here, using multigene and genome-based data, we assemble a 5,284-species phylogenetic tree and infer ages and broad patterns of speciation/extinction and morphological innovation in mushroom-

forming fungi. Agaricomycetes started a rapid class-wide radiation in the Jurassic, coinciding with the spread of (sub) tropical coniferous forests and a warming climate. (Varga *et al.* 2019)

Mushrooms with enhanced medicinal properties focus on finding such compounds that could modulate the human body's immune systems. Mushrooms have antimicrobial, antidiabetic, antiviral, hepatoprotective, antitumor, and immunomodulatory properties due to the presence of various bioactive components. β-glucans are the major constituent of the mushroom cell wall and play a significant role in their biological activity. This review described the techniques used in the extraction of the active ingredients from the mushroom. We highlighted the structure of the bioactive polysaccharides present in the mushrooms. Therapeutic applications of different mushrooms were also described. It is interesting to note that mushrooms have the potential sources of many bioactive products that can regulate immunity. Thus, the development of functional medicinal food based on the mushroom is vital for human welfare.(Patel *et al.* 2021)

Mushroom, a fungi kingdom member, has drawn significant interest in medicinal applications due to their anti-inflammatory, antimicrobial, antidiabetic, cardiovascular-protective, hepatoprotective, and anticancer potentials. It regulated the immune system, and macrophages, T cells, dendritic cells (DC), natural killer (NK) cells, and hematopoietic stem cell activities by phagocytic activity, generating reactive oxygen species, inflammatory mediators, and cytokines production. Mushroom is a rich source of proteins, fibers, and vitamins. It is challenging to distinguish between edible and medicinal mushrooms because many edible mushrooms have therapeutic properties, and many medicinal mushrooms are also edible. Approximately 1.5–5.0 million fungi species are known globally, and among them, a few hundred species have medicinal properties(Patel *et al.* 2021)

Mushroom polysaccharides are a type of bioactive macromolecular which isolated from fruiting bodies, mycelia or fermentation broths of edible or medicinal fungus. Recently, mushroom polysaccharides have attracted a lot of attention for regulating gut microbiota via reducing the levels of pathogens and stimulating the growth of beneficial microorganisms, thus creating new possibilities for their use in nutraceutical and functional foods industries. The current review summarizes the isolation, purification and structural characterization methods of mushroom polysaccharides, the degradation of mushroom polysaccharides in intestine, the impacts of mushroom polysaccharides on gut microbiota community and short chain fatty acids (SCFAs)

productivity, and the beneficial effects of mushroom polysaccharides to host by targeting gut microbiota.(Yin *et al.* 2020)

The use of spent mushroom substrate (SMS) in new cultivation cycles has already been reported due to its economic and environmental viability. When considering the application of the circular economy concept in the production of edible mushrooms, the re-use of the SMS within the same process is highly attractive, because it allows a better use of the biomass and the energy involved in the process and, therefore, tends to improve energy efficiency and resource conservation. However, this alternative generates important challenges, which derive from maintaining the quality standards of the mushrooms produced and, at the same time, not incurring excessive costs that are detrimental to the process itself. In our opinion, the main difficulty of the process in achieving success is regarding the biological and agronomic parameters that involve the production of the mushroom. It is useless to apply SMS in new cycles if the mushroom harvest is impaired and farms become non-viable. (Cunha Zied *et al.* 2020)

To date, 150,000–160,000 mushrooms have been recorded, however, among them, only about 100 species of the prime edible mushrooms are economically cultivated, around 60 commercially cultivated are widely accepted as food sources, and more than 10 produced on an industrial scale in many countries. Most edible mushrooms are rich in nutrients, such as protein, polysaccharides, vitamins, mineral elements as well as various functional factors, all substances are essential for human, as a result, long-term consumption will have a beneficial effect on improvement of human health. Generally speaking, fresh mushrooms contain 85–90% moisture, and the mentioned nutritional values is recognized as components in dried individuals.(Gong *et al.* 2020)

Even though mushrooms are often considered to be a vegetable, they actually belong to the fungal kingdom. There are >2000 species of mushrooms in nature, ≥25 of which are widely accepted as functional foods for human consumption and commercially cultivated. Overall, mushrooms have been reported to have anticancer capabilities and protective effects against tumor development and laboratory studies have revealed these anticarcinogenic effects vary according to different types of mushrooms such as shiitake, maitake, and *Agaricus bisporus* (button mushroom). Several epidemiological observational studies have also reported an inverse association between mushroom consumption and cancer risk. However, several other

epidemiological studies that have examined the effects of mushroom intake on the risk of cancer have yielded nonsignificant associations. (Ba *et al.* 2021)

Mushrooms belong to phylum basidiomycetes which are huge and amazingly diverse group that along with the ascomycetes, constitute the kingdom *Dikarya*, collectively referred to as the "higher fungi" (Shipley et al., 2016). They have a distinctive body that be can be epigeous or hypogeous, large enough that they are visible to the naked eye and can be picked by hand. Majority of the edible mushroom species belong to the class *Agaricomycetes* that produces basidiocarps such as gilled mushrooms, puffballs, bracket, coral, jelly and crust fungi (Hibbett, 2006). There are about 12,000 species of such mushrooms available worldwide. Of these, 2000 species are considered edible and 35 are commercially cultivated throughout the world (Rathore, Prasad, & Sharma, 2017). Further, mushrooms are known to possess extensive enzyme complexes that enable them to flourish successfully on lignocellulosic substrates containing lignin, celluloses, hemicelluloses and pectin (Rathore *et al.* 2019)

In the developed world communicable diseases account for maximum proportion of ailments nowadays. Global morbidity and even mortality are sometimes associated with common pathogens. Several severe infections and complications occur in the present context such as acute gastroenteritis (AGE), colitis, bloodstream infection, meningitis, joint infection, strep throat, urinary tract infections (UTIs), bacterial food poisoning, bacterial cellulitis and vaginosis, gonorrhea, chlamydia, syphilis, respiratory tract infection etc. The extensive use of antibiotics and fungicides to protect against infections by pathogens contributed immensely to many medical achievements. On the downside of these measures a high number of multi-drug resistant pathogens has also emerged of late(Pandey *et al.* 2020)

Dramatic increases in the speed and quality of imaging, segmentation and reconstruction in electron microscopy now allow large-scale, dense connectomic studies of nervous systems. Such studies can reveal the chemical synapses between all neurons, generating a complete connectivity map. Connectomics is particularly useful in generating biological insights when applied to an ensemble of neurons with interesting behavioral functions that have already been extensively studied experimentally. Knowing the effects on behavior and physiology of perturbing individual cell types that can also be unambiguously identified in the connectome is of

considerable value. Here, we present a connectomic analysis of one such neuronal ensemble, the mushroom body (MB) of an adult *Drosophila melanogaster*.(Li *et al.* 2020)

Mushroom substrate is a mushroom growing media that consists of several organic materials such as sawdust and rice bran. The substrate media can no longer be used and needs to be disposed of upon completion of several cycles of mushroom harvesting. The production of mushrooms in Malaysia was estimated to be 24 ton day^{-1}. This led to generation up to 5 kg of by-product (i.e. spent mushroom substrate (SMS)) from 1 kg of mushroom product. Therefore, approximately 120 metric tons of SMS have been produced daily by the mushroom industry.(Lam *et al.* 2019)

Lignocellulosic biomass with inherent recalcitrance embedded in the structural components has valuable and sustainable energy sources to produce environmentally friendly energy, and the use of that energy plays a crucial role in sustainable development strategies. Lignocellulosic biomass is considered a good source of renewable feedstocks and contains high fermentable sugars. However, the lignocellulosic-based feedstock is a recalcitrant material that requires intensive labor and high capital cost for processing. Biodegradation of the lignocellulosic materials is associated with harsh processing conditions such as extreme pH and temperature, high salinity, metal ions, and organic solvents; therefore, laccases can function in these harsh conditions are in increasing demand. Incorporating laccases in the bioconversion of lignocellulosic biomass is very attractive and advantageous due to the depolymerization of lignin under a milder and cleaner conditions without damaging the cellulose . These ligninolytic enzymes are considered ecofriendly alternatives for biodelignifying natural wastes offering economic benefits through decreasing lignocellulosic biowastes accumulation. Bacteria can effectively degrade lignocellulose.(Gałązka *et al.* 2023)

Lignocellulosic biomass is composed of three major biopolymers: cellulose, hemicellulose and lignin. Analytical tools capable of quickly detecting both glycan and lignin deconstruction are needed to support the development and characterization of efficient enzymes/enzyme cocktails. Previously we have described nanostructure-initiator mass spectrometry-based assays for the analysis of glycosyl hydrolase and most recently an assay for lignin modifying enzymes. Here we integrate these two assays into a single multiplexed assay against both classes of enzymes and use it to characterize crude commercial enzyme mixtures. Application of our multiplexed platform

based on nanostructure-initiator mass spectrometry enabled us to characterize crude mixtures of laccase enzymes from fungi *Agaricus bisporus* (*Ab*) and *Myceliopthora thermophila* (*Mt*) revealing activity on both carbohydrate and aromatic substrates. Using time-series analysis we determined that crude laccase from *Ab* has the higher GH activity and that laccase from *Mt* has the higher activity against our lignin model compound. Inhibitor studies showed a significant reduction in Mt GH activity under low oxygen conditions and increased activities in the presence of vanillin (common GH inhibitor). Ultimately, this assay can help to discover mixtures of enzymes that could be incorporated into biomass pretreatments to deconstruct diverse components of lignocellulosic biomass.(Ghorai 2022)

There is a growing need to find renewable and eco-friendly sources of energy. Lignocellulosic biomass has great potential to create sustainable drop-in replacements for conventional petroleum-based fuels. It is composed of three major biopolymers: cellulose, hemicellulose and lignin and therefore maximal breakdown will require efficient deconstruction of both the glycans via glycosyl hydrolases (GHs) and lignin by lignin modifying enzymes (LMEs). It is desirable to develop chemically resolved assays that are suitable for simultaneously characterizing both GH and LME activities because lignin products can impede enzymatic saccharification through nonproductive binding interactions with carbohydrate-active enzymes. Finding enzymes/enzyme cocktails which can cleave both carbohydrate and phenolic bonds has great potential to help prevent this deactivation and aid in both lignin and glycan deconstruction towards the development of bioproducts.(Giannakopoulou *et al.*)

Laccases are a class of lignin-active enzymes which catalyze single-electron oxidations of aromatic substrates by using oxygen as a terminal electron acceptor. When coupled with a mediator such as 2,2'-azinobis(3-ethylbenzathiazoline-6-sulfonate) (ABTS), laccases can oxidize nonphenolic substrates as well, making them a target enzyme for lignin deconstruction. To our knowledge no specific analytical methods have been developed to simultaneously characterize the breakdown product distribution of lignin and cellulose model compounds.Here, we report the integration of existing NIMS assays using cellobiose probes with the recently reported phenolic β-aryl ether probe to simultaneously assay activities against lignin and glycan model compounds. We selected the β-aryl ether bond for our studies because it is the most common linkage in lignin.

Epigallocatechin-3-O-gallate (EGCG), one of the most abundant monomeric catechin compounds in green tea, was oxidized to EGCG dimers through radical oxidative reaction using laccase, a member of the tea polyphenol oxidase family. Electrospray ionization tandem LC-MS (ESI-LC/MS) was applied for the characterization of the dimers, including theacitrin, dehydrotheasinensin, and theasinensin, which were produced by the radical oxidative reaction. Chemical structures of the EGCG dimer were identified using LC/MS/MS and fragmentation patterns of EGCG dimer. The dimers were isomerized by heat treatment at 80°C after oxidation step to mimic heat treatment and drying in the last step of leaf fermentation. As a result of the heat treatment, theasinensin A (TSA) was formed as a major dimer product. After the two steps of Sephadex LH-20 and C18 column chromatography, 27.6 mg of purified TSA was obtained from 485.4 mg of dried reaction mixture. Biological effect of TSA as antioxidant and anti-aging effects were investigated. (Giannakopoulou *et al.* 2022)

Postharvest browning in white button mushroom is a major limiting factor determining the quality and marketability of mushrooms. In the present study, 361 hybrids were developed using non-fertile isolates from eleven button mushroom strains. A total of forty one browning tolerant hybrids were selected and evaluated for the yield. Five of these were evaluated on large scale in commercial units and two have been finally selected on the basis of higher yield than the strains used by the commercial units. The selected five hybrids showed very low level of enzymes responsible for browning i.e., laccase, polyphenol oxidases and tyrosinase.The hybrids were characterized using SSR, IRAP and ReMAP markers and distinguishing bands for hybrids could be identified. The developed hybrids will address postharvest browning problems of button mushroom industry.(GUGALIA)

Marine organisms, especially macroalgae, constitute a renowned source of bioactive compounds. Among marine macroalgae, *Ulva* species, green seaweeds belonging to the order Ulvales (Chlorophyta), have recently attracted scientific interest. Specifically, *Ulva intestinalis*, a bright grass-green seaweed, rich in polysaccharides, proteins, minerals, and other bioactive substances, such as phenolics has gained increasing attention over the last years. These species produce a wide variety of secondary metabolites with a broad array of bioactivities, including antitumoral, antiviral, antifungal, antibacterial, antiaging, cytotoxic and antiproliferative activity.

For example, a protein fraction from *Ulva intestinalis* induced a remarkable increase in collagen and hyaluronic acid production per cell and reduced cell proliferation without increasing cell mortality. Although these species are yet much underexploited, the remarkable potential arising from their exploitation approves their compelling scientific interest, especially in the pharmaceutical and food industries.(Hidalgo *et al.* 2023)

The blue multicopper oxidase enzyme laccase (EC 1.10.3.2) has the ability to catalyze the oxidation process of phenolic and non-phenolic aromatic substrates while also reducing molecular oxygen to water. Using molecular oxygen as an electron acceptor, laccase oxidizes its substrates by an electron transfer mechanism that generates unstable free radical intermediates and causes non-enzymatic reactions that break down substrate molecules . Laccase substrate specificity differs from one organism to another, and in the presence of adequate redox mediators, the range of laccase oxidizable substrates can be greatly extended. Laccase is a glycosylated monomer with a molecular mass ranging from 54 to 97 kDa, relying on the source species. Numerous environmental pollutants, such as bisphenol A (BPA), may be successfully broken down and mineralized by fungi using their laccases. Laccase belongs to the polyphenol oxidases family, and its active site contains four copper atoms from types T_1, T_2, and T_3.(Iqhrammullah *et al*. 2023)

Laccase enzymes, multi-copper oxidases found in plants, insects, fungi, and bacteria, have inspired biobased strategies for the treatment of recalcitrant compounds due to their ability to oxidatively degrade lignin using molecular oxygen as a terminal electron acceptor. Bacterial laccases tend to have a higher thermotolerance and a wider pH range, while fungal laccases tend to have higher redox potentials (~0.5–0.8 V vs NHE). While they exhibit broad substrate specificity, the performance of laccase with respect to contaminant transformation is limited by its oxidation potential. (Iqhrammullah *et al.* 2023)

In the current scenario 64 %, 26 %, and 10 % ethanol is produced from maize, sugarcane, and wheat flour or other grain sources respectively by the traditional fermentation process. In 2020, the United States of America was the leading country to produce the highest (13.8 billion gallons) amount of ethanol; Brazil was the second-largest alcohol producer (7.9 billion gallons), while 0.88 and 0.48 billion gallons were produced by China and India respectively. Recently it was estimated; globally ethanol generation will increase up to 37 billion gallons by 2029. As a result, to expect

the production of such a huge amount of alcohol will be needed of >25 % and 14 % of global sugarcane and maize production respectively (Agricultural outlook OECD/FAO, 2020). Such a diversion and misuse of food grains to generate alcohol are considered as one of the main reasons to raise food prices all over the world. On the other hand, corn and many other grains are also used as feedstock for poultry and in livestock management as a part of the food based economy.(Jayakumar *et al.* 2023)

Global threat from emerging contaminants to human health has pushed researchers to develop pollutant removal technologies including those based on biocatalytic degradation. Laccase, a multicopper enzyme, is reported for its applicability in treating wastewater and industrial effluent by catalyzing reduction of O_2 to H_2O in the presence of substrates (phenolic or non-phenolic compounds). Scientific papers have discussed the ability of this enzyme to breakdown hazardous contaminants ranging from azo dyes, antibiotics, and up to hormones. Nonetheless, the cost production of this enzyme is high, hence the importance of the immobilization into solid supports. Polymers are considered suitable for the immobilization because they are easy for modification and have adjustable characteristics. This review discusses the immobilization of laccase into polymers via various techniques (adsorption, cross-linking, covalent binding, encapsulation, and entrapment) and its implications on the pollutant removal performance. In addition, bibliometric analysis is performed and summarized the advantages and disadvantages of each immobilization technique.(Johnnie *et al.*)

Increasing anthropogenic activities have contributed to the massive pollutants discharge into environment ascribed to insufficient wastewater infrastructure and limited options for disposal methods.These conditions particularly become more challenging in the face of coronavirus disease 2019 (COVID-19) pandemic, where organic waste is discharged to rivers and water bodies without proper treatment.Excessive and inefficient use of cleaning and sanitizing agents during the pandemic has alarmed researchers regarding their deleterious impact on environment.On daily basis, humans and other organisms are threatened by the presence of emerging contaminants which include pharmaceuticals (antibiotics, hormones, etc.), pesticides and herbicides, plastic additives, non-halogenated compounds, and so on. Non-nutritive sweetener, a sugar substitute used in

various food products, is also considered an emerging contaminant owing to its resistance and stability in aquatic environments.(Juan *et al.* 2022)

One-dimensional Ni/Au/PPy-COOH nanowires with multiple segments were synthesized in this study. Smooth surfaces and magnetic properties of nanowires were investigated by scanning transmission electron microscopy (SEM), Energy-dispersive X-ray spectroscopy (EDX), and Electron Spin Resonance (ESR) techniques. The nanowires were used to modify the screen-printed electrode surface and as a micro-environment for *Trametes versicolor* laccase. The ability of this enzyme biosensor to detect dopamine change in human biological samples was demonstrated by a wide linear range (0.01–50 μM) and a low LOD (2.265 nM). In addition, the biosensor exhibited excellent selectivity allowing the detection of dopamine in the presence of ascorbic acid, uric acid, L-Cys, serotonin, and glucose, with high sensitivity of reduction currents obtained at −0.2 V (*vs.* Ag/AgCl). The proposed biosensor allowed the detection of dopamine in commercial serum and artificial urine with recovery values close to 100 %. It also demonstrated reproducibility, reusability, and long-term storage stability. The sensitivity, K_m^{app}, and I_{max} values of the biosensor were determined as 2.05 μM and 1.03 μA, respectively. The LAC-Ni/Au/PPy-COOH/NAF/SPE biosensor is a reliable design for detecting dopamine with a wide linear range.(Kaur *et al.* 2022)

Dopamine (DA), 3,4-dihydroxyphenethylamine, is a catecholamine neurotransmitter in the transmission between cells and is found in the human body. Dopamine is a primary amine belonging to the biogenic amine class and plays a significant role in the functioning of the central nervous, renal, hormonal, cardiovascular, and endocrine systems. High levels of dopamine in the body are an indicator of diseases such as rapid heart rates, hypertension, and heart failure. While it is thought that low dopamine levels may cause neurological disorders such as Parkinson's disease, Schizophrenia, Alzheimer's disease, stress, and depression (the dopamine level in a healthy individual is in the range of 0.01–0.1 μM). Dopamine measurement is usually performed using analytical techniques such as liquid chromatography and capillary electrophoresis. Despite their low detection limits, these techniques have significant limitations, such as large equipment requirements, long analysis times, and cause organic waste. For these reasons, it is pivotal to develop biosensors for faster and more convenient detection of dopamine as a biomarker. The co-existence of small molecules such as ascorbic acid and uric acid in the blood, urine, and

extracellular fluids causes high interference at similar potentials on bare electrodes during the detection of dopamine.(Kumar *et al.* 2022)

A multicopper oxidase, laccases catalyze the four-electron reduction of the substrate with the use of molecular oxygen. Laccases are abundant in nature and can be found in virtually every form of life on the planet. Generally speaking, laccases are classified into three types: blue, white, and yellow. Plant, bacterial and fungal laccases all have the same trinuclear copper site for substrate reduction. Non-phenolic as well as phenolic molecules are both capable of being catalyzed by this enzyme. Laccases are used in a wide range of industries that make use of phenolic chemicals. Laccases have been the subject of recent research because of their unique features. Laccase, its sources, manufacture, purification, and applications in many sectors are discussed in length in this review.(Kyomuhimbo & Brink 2023)

The search for ecologically friendly technologies has increased interest in using enzymes to replace conventional non-biological methods, which are increasingly becoming popular. Laccases (benzenediol: oxygen oxidoreductases; EC 1.10.3.2) have received the most attention in recent decades when comparing the many oxidant enzymes currently exist. In nature, laccases are multicopper oxidases that may be found in abundance. They catalyze the one-electron oxidation of phenolic compounds, resulting in the simultaneous reduction of oxygen to water in the presence of oxygen.(Leutbecher *et al.* 2009)

Enzymes are utilized in industrial processes to make them cleaner and more effective, as well as to support the idea of sustainable development. Enzymatic reactions have a number of advantages over conventional chemical processes, which leads to improved resource utilization and reducing waste in product development. In this aspect, laccases (EC.1.10.3.2) are metals containing mononuclear enzymes which can oxidize a wide range of phenols, aromatic amines, and heterocyclic compounds with low redox potential properties. According to the UniProtKB database search results for "laccase," the enzyme is identified in 1026 bacteria, 6258 eukaryotes, and 16 halobacteria (archaea), with sequence sizes ranging from 220 to 800 amino acids. As a result, it is reasonable to expect that the incredible number of enzymes generated by various species will have a wide variety of uses. (Leutbecher *et al*. 2009)

Green chemistry elements should deal with enzymatic chemical reactions that are efficient or have a low environmental impact. Laccase is a readily available multicopper oxidase that has been extensively studied for a long time. In reviewing recent trends in laccase research, the authors selected themes that have not yet been evaluated, targeting open-access journals. From a chemical viewpoint, decomposition reactions (such as oxidation of substrates, toxic substances, pigments, and biomass materials), organic synthesis reactions, composite catalytic activities of oxygen reduction reactions, biosensors, batteries, biological fields, fermentation, culture, gene association, and antifungal activity.(Li *et al.* 2022)

The growing demand for biofuels production using vegetable oils leads to a continuous increase of the amount of glycerol, the secondary product of the biodiesel industry, available on the market. Even though efficient large-scale recovery and valorization methods of this low-cost byproduct are still being developed, there is an extensive research focusing on this topic. Since the volumetric biodiesel to glycerol ratio is 10:1, the continuously increasing production of this biofuel was estimated to generate globally 7.66 million tons glycerol in 2020 and, therefore, a consistent polyol surplus on the market, with the consequent need of novel sustainable technologies for its conversion in value-added products. Glycerol (1,2,3-propanetriol) is an important starting material for the synthesis of many important chemical compounds by various catalytic routes such as pyrolysis, reforming, hydrogenation, oxidation, or esterification, subject of several biorefinery strategies.(Li *et al.* 2022)

Plant phenolic compounds demonstrate bioactive properties in vitro and/or in vivo, which creates demand for their precise determination in life sciences and industry. Measuring the concentration of individual phenolic compounds is a complex task, since approximately 9000 plant phenolic substances have been identified so far. The determination of the total phenolic content (TPC) is less laborious and is used for the qualimetric evaluation of complex multicomponent samples in routine analyses. Biosensors based on phenol oxidases (POs) have been proposed as alternative analytical devices for detecting phenolic compounds; however, their effectiveness in the analysis of food and vegetal matrices has not been addressed in detail. This review describes catalytic properties of laccase and tyrosinase and reports on the enzymatic and bienzymatic sensors based on laccase and tyrosinase for estimating the total phenolic index (TPI) in food-related

samples (FRSs). The review presents the classification of biosensors, POs immobilization, the functions of nanomaterials, the biosensing catalytic cycle, interference, validation, and some other aspects related to TPI assessment. Nanomaterials are involved in the processes of immobilization, electron transfer, signal formation, and amplification, and they improve the performance of PO-based biosensors. Possible strategies for reducing interference in PO-based biosensors are discussed, namely the removal of ascorbic acid and the use of highly purified enzymes.(Lim & Mohamad 2022)

Laccase is a key enzyme in the fungal 1,8-dihydroxynaphthalene(DHN) melanin biosynthesis pathway, which is a potential target for the control of pathogenic fungi. In our previous work, **a2** was found with higher inhibition activity against laccase and antifungal activity than laccase inhibitor PMDD-5Y. The introduction of hydrogen-bonded receptors in amino part was found to be beneficial in improving laccase inhibitory activity by target-based-biological rational design. In this work, the hydrogen-bonded receptors morpholine and piperazine were introduced for structure optimization to enhancing biological activity. Enzyme activity tests indicated that all target compounds had inhibitory activity against laccase, and some compounds exhibited better activity against laccase than **a2**, it was further verified the introduction of hydrogen-bonded receptors in the amino portion could enhance the laccase inhibitory activity of target compounds. Most compounds showed excellent antifungal activities *in vitro*.(Lin *et al.* 2023)

Regarding inorganic contaminants, Raj et al. (2011) highlighted the capacity of some fungi to accumulate high concentrations of metals (e.g., Pb) in their fruiting bodies. Furthermore, Kulshreshtha (2018) and García-Delgado et al. (2013) reported that SMS can act as metal adsorbent due to the presence of fungal mycelia. Nevertheless, there is still insufficient knowledge on the suitability of SMS to reduce bioavailable metal concentrations in soil. In any case, mycoremediation is widely recognized as a bioremediation strategy of great potential, particularly for soils contaminated with recalcitrant organic compounds (Kulshreshtha et al., 2014; Singh, 2006). Importantly, the use of SMS for mycoremediation purposes (an example of waste valorisation) is in accordance with the circular economy paradigm.(Liu *et al.* 2022).

Aims and Objectives

1. Growth optimization and cultivation of *Agaricus bisporus* (white button mushroom) on cheaper substrate.
2. Structure prediction, validation, and comparative analysis of *A. bisporus* Laccase (AbL) predicted model.
3. Docking of Glycerol (GLU) in AbL predicted model to highlight the interactions between enzyme and ligand.